Annotation Convention, Not Architecture, Limits Cross-Facility Detection of Electric Vehicle Battery Components

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Abstract: Automated disassembly of end-of-life electric vehicle (EV) battery packs depends on a detector that locates modules and busbars reliably across the pack types a recycling line actually receives. Published detectors for this task are trained and validated on imagery from a single facility, and the accuracy they report is shown here to be a poor predictor of behaviour elsewhere. A cross-facility benchmark of 225 images and 780 instances, assembled from 13 independent public sources spanning ten EV pack families and verified free of train-test leakage by perceptual hashing, is presented. On this benchmark a single-source specialist falls from 0.818 to 0.277 mAP@50, a relative loss of 66%. Disaggregating accuracy by source reveals that the loss is not distributed evenly: per-source module accuracy spans 0.995 to 0.043, and the collapse occurs on exactly one source whose annotation convention differs from the consensus. This observation is developed into a screening procedure that identifies convention-divergent sources from a single merged-corpus training run using a robust outlier criterion on per-source accuracy. Applying the proposed procedure and excluding the one flagged source raises module mAP@50 from 0.547 to 0.774 without additional data or training budget. Two architectures are compared under an identical evaluation protocol. A detection transformer built on a frozen self-supervised backbone confers no advantage on convention-consistent sources, at 0.771 against 0.774, but degrades far more gracefully when the annotation convention shifts, at 0.680 against 0.547, which isolates robustness to annotation shift rather than detection accuracy as the property such backbones supply. A training-free variant of the screen, computed from annotation geometry alone over 4,760 label files, flags the same source with no training run, giving a second and methodologically independent confirmation, and passes a positive control in which a consensus source relabelled to a sub-component convention is flagged while every genuine consensus source is not. Four negative results are reported that constrain the design space. The benchmark, the per-source provenance record and the evaluation code are released.

1 INTRODUCTION

Electric vehicles are reaching end of life (EoL) in large numbers, and the battery packs they carry are simultaneously a hazard and a source of critical raw materials. Lithium, cobalt and nickel are all classified by the European Commission as critical raw materials with elevated supply risk (European Commission, 2023; Harper et al., 2019), and European recycling capacity demand is projected to reach 170 GWh by 2035 (Transport and Environment, 2024). Regulation (EU) 2023/1542 additionally requires documented condition data across the battery lifecycle (European Parliament and Council, 2023), which manual inspection delivers inconsistently.

Disassembly of EoL packs nevertheless remains predominantly manual. The operation is labour intensive, slow, and exposes workers to high voltage and to the risk of thermal runaway. A substantial research effort is consequently directed at automating it (Zang et al., 2024; Tan et al., 2025).

Every such system begins with perception. Before a manipulator can unfasten, grasp or lift a component, the battery modules and the busbars that connect them must be located in the image. The established approach is a supervised object detector trained on imagery collected at a single facility or from a single experimental rig. This practice conceals a problem specific to the application. A recycling line does not receive a single pack type; it receives whatever arrives on a given day, from different manufacturers, in different states of disassembly, under whatever illumination the building provides. Accuracy measured on the distribution a detector was trained on therefore carries limited information about the next pack through the door.

To quantify this effect a cross-facility benchmark is required, and none exists for the task. Every study surveyed in Section 2 collected and annotated its own imagery, and none released it. To address this gap, this paper assembles a corpus of 4,425 training images from 13 independent public sources spanning ten EV pack families, constructs a held-out cross-facility benchmark of 225 images containing 780 instances, and evaluates detectors on it under a single harmonised protocol.

The principal finding is that the dominant failure mode in this setting is neither image difficulty nor model capacity. It is disagreement between sources about what the target class denotes. Per-source module accuracy across the benchmark spans 0.995 to 0.043, and inspection confirms that the source on which accuracy collapses annotates small internal subcomponents as modules while every other source annotates pack-level modules. The detector learned the majority convention and applied it consistently, so the collapse constitutes a disagreement about the target rather than a failure of perception.

This observation is developed into a screening procedure. The contributions of this paper are as follows:

1. A cross-facility benchmark for EV battery module and busbar detection, comprising 225 images and 780 instances drawn from 13 sources and ten pack families, verified free of train-test leakage by perceptual hashing and released with a per-source provenance and licence record.

2. A convention-consistency screening procedure that identifies annotation-divergent sources from a single merged-corpus training run using a robust outlier criterion on per-source disaggregated accuracy. Excluding the one source flagged by the proposed procedure raises module mAP@50 from 0.547 to 0.774, with no additional data and no additional training budget.

3. Quantification of the single-source evaluation gap for this task: 0.818 mAP@50 in domain against 0.277 across facilities, a relative loss of 66%.

4. A characterisation of what a frozen self-supervised backbone supplies in this setting. No advantage is observed on convention-consistent sources, at 0.771 against 0.774 on the module class; the entire aggregate advantage arises from graceful degradation under annotation shift.

5. A training-free variant of the screening procedure, operating on annotation geometry alone, which flags the same source without any training run and therefore permits a corpus to be screened before a detector is trained.

6. Four negative results, covering open-vocabulary detection, automatic relabelling, naive multi-source merging and synthetic augmentation, each of which constrains the design space for this task.

Relation to prior work by the same authors. The single-source specialist evaluated in this paper is the detector reported in (Katiyar et al., 2026), which addressed detection and condition assessment at a single facility using a CPU-deployable two-stage pipeline. That work could not test cross-facility transfer, which is precisely the limitation the present paper addresses. The corpus, the benchmark, both screening procedures and all evaluation reported here are new, and the condition-assessment stage of the earlier work is deliberately not revisited.

2 RELATED WORK

2.1 Perception for Battery Disassembly

Zang et al. (2024) surveyed robotic EV battery disassembly and concluded that the process remains predominantly manual, with perception and task uncertainty identified as the principal obstacles. Wegener et al. (2014) demonstrated that pack designs vary sufficiently between manufacturers that fixed automation does not transfer, which is the practical reason perception must generalise. However, that study addressed mechanical disassembly sequencing and did not evaluate a perception component.

Most published perception work for this application targets fasteners rather than components. Li et al. (2021) proposed a screw detection method combining Faster R-CNN with rotation edge similarity, reporting improved localisation of the screw axis. However, the method was validated on a single collection of laboratory imagery. Li et al. (2023) developed a YOLOX-family detector for active screw detection in EV battery disassembly using 1,719 hand-annotated images. However, the dataset was not released and accuracy was reported only in domain. Naseri et al. (2026) presented a comparison of YOLOv11 against Faster R-CNN for the same task, and observed that the field is hampered by a lack of public datasets; the study resorts to laptop screws as a proxy domain, which limits the conclusions that transfer to battery imagery.

Module-level and busbar-level perception is comparatively under-explored. Tan et al. (2025) developed a collaborative robot that unfastens and collects screws by fusing vision with force and torque sensing, reporting above 90% unfastening efficiency. However, the vision component operates on a single fixed pack geometry, so the reported figure does not characterise transfer. Gerlitz et al. (2023) applied instance segmentation with point-cloud registration in order to grasp busbars on an industrial module. However, the approach is restricted to one module type and requires a reference model of that type.

A recurring theme is data scarcity. Every study named above collected and hand-annotated its own imagery, and none released it. That absence motivates the benchmark presented in this paper.

2.2 Cross-Dataset Evaluation

Domain shift is a well-documented failure mode in visual recognition. Recht et al. (2019) demonstrated that classifiers which appear strong on the original ImageNet test split lose accuracy on a replication set collected by the same protocol. However, that study varied the sampling procedure while holding the label taxonomy fixed. Koh et al. (2021) assembled the WILDS benchmark of in-the-wild distribution shifts across ten application domains, and established that standard training degrades substantially under such shift. However, WILDS likewise holds the annotation protocol constant within each dataset, whereas the corpus considered in this paper varies both the imaging distribution and the annotation protocol between sources.

Northcutt et al. (2021) demonstrated that pervasive label errors in established test sets destabilise benchmark rankings, and estimated an average label error rate of 3.4% across ten widely used datasets. However, that work treats label error as noise about an agreed target, whereas the failure characterised here is systematic disagreement about what the target is. The distinction is consequential for merged corpora: noise averages out with scale whereas a divergent convention does not, and the proposed screening procedure operationalises exactly that distinction.

Penquitt et al. (2026) developed Rechecked, a modular framework combining automatic label-error detection with crowd-sourced verification, and applied it to the pedestrian class of KITTI, in which 18% of the original ground-truth labels were found to be missing or inaccurate. However, the framework corrects errors relative to an annotation standard that is agreed within a single dataset, and the authors report that the best available detection methods still miss up to 66% of label errors. The failure characterised in the present work is of a different kind: the sources do not disagree about individual boxes, they disagree about what the class denotes, and consequently no correction procedure operating within a single convention can reconcile them. Section 5.8 reports an attempt at exactly such a reconciliation, and its failure.

Abou Akar et al. (2026) investigated multi-source training for industrial object detection, combining real imagery with rendered and generatively synthesised data, and demonstrated that the composition of the training mixture materially affects generalisation. However, the sources compared in that study share a single annotation standard by construction, because the synthetic data is generated with ground truth attached. The corpus considered here is multi-source in a stricter sense, in that each contributing source carries an independently authored annotation standard.

2.3 Self-Supervised Representations for Detection

Self-supervised backbones such as DINOv2 (Oquab et al., 2024) learn features that transfer without task-specific fine-tuning, and detection transformers (Carion et al., 2020) built on frozen variants of them (Roboflow, 2025) are reported to generalise better than detectors trained from scratch in low-data regimes. However, that claim has been established principally on natural imagery with a consistent annotation protocol. Whether it holds on the cluttered, specular imagery of battery internals, and whether it holds under annotation shift specifically, is evaluated in Section 5.5.

2.4 Open-Vocabulary Detection and Automatic Annotation

Open-vocabulary detectors such as YOLO-World (Cheng et al., 2024) and Grounding DINO (Liu et al., 2024) localise objects specified by free-text prompts and have been proposed as a route to annotation-free deployment and to automatic pre-annotation. However, their accuracy depends on the target concept

Table 1: Principal corpus sources. Thirteen sources in total, spanning ten EV pack families.

Source	Images	Pack family
ev-battery-comp-det.	1,045	multi-manufacturer
ev-battery-components	680	Audi / BMW / Chevrolet
last_exp4	483	multi-manufacturer
ev-battery-pack	435	single laboratory rig
battery_comp	280	Nissan Leaf
ue_d1_defect	237	module close-ups
final_mobilenet	216	multi-manufacturer
automated-disassembly	117	disassembly cell
bmw_i3	111	BMW i3
ue_rav4	96	Toyota RAV4
Others (three sources)	725	mixed

being represented in the vision-language pre-training corpus. The behaviour of both on battery internals is reported in Section 5.8.

Benchmark design for industrial vision has precedent outside this application. Bergmann et al. (2019) introduced MVTec AD, which established that a carefully constructed benchmark with documented acquisition conditions can redirect an entire sub-field. However, MVTec AD was collected by one group under controlled conditions and therefore does not exhibit the inter-source annotation disagreement examined here. Zhou et al. (2023) surveyed domain generalisation methods and organised them into data manipulation, representation learning and learning strategy. However, that taxonomy presumes a fixed label space across domains, which is the assumption the present work shows to fail on assembled corpora.

To the best of our knowledge, no public cross-facility benchmark for EV battery component detection exists, and no prior study has quantified the contribution of annotation-convention divergence to cross-source detection accuracy. Therefore, this research presents both.

3 CORPUS AND BENCHMARK

3.1 Sources

A corpus of 4,425 training images was assembled from 13 independent public sources spanning ten EV pack families, annotated with two classes, module and busbar. Table 1 presents the principal sources with their image counts and pack families.

The images and the polygon annotations originate with those sources. The contribution made in this paper is the assembly, the deduplication, the audit and the benchmark construction, not the annotation itself. Licences differ across sources and one is non-commercial, so per-source provenance is recorded and released so that any single source can be excluded by a downstream user.

Table 2: Annotation and image-quality audit of the 4,425-image training corpus.

Flag	Share	Assessment
Unlabelled	23.5%	partly harmful
Low blur score	12.3%	benign
Greyscale	20.4%	benign
Degenerate boxes	0.0%	—
Polygon annotations	86.8%	16,945 of 19,522

3.2 Deduplication and Leakage Control

Reported accuracy is informative only against a benchmark that does not overlap the training data. All 4,440 candidate images were therefore compared by difference hashing (dHash) at a Hamming distance threshold of 6. That threshold was selected because it separates augmentation-derived near-duplicates from genuinely distinct frames on this corpus; values below 4 admitted rotated duplicates and values above 8 rejected distinct frames of the same pack.

Fifteen same-source near-duplicates were removed. A subsequent cross-source pass at a looser threshold returned zero duplicates, which confirms that the sources are genuinely independent collections rather than republications of one another.

This control proved necessary rather than precautionary. One candidate source, a 712-image public collection covering 19 vehicle types, republishes imagery from another source already present in the corpus: 78 of its images were byte-identical duplicates of training images, a fact stated in its own documentation but easily overlooked. Merging without verification would have placed training images directly into the benchmark and inflated every subsequent measurement.

3.3 Annotation Audit

All 4,425 training images were audited for label and image quality. Table 2 presents the outcome. Of the corpus, 23.5% carries no labels, 12.3% scores low on a Laplacian-variance blur metric, 20.4% is greyscale, and no image contains a degenerate box.

Two findings changed how the corpus was treated. Firstly, the blur and greyscale flags are largely false alarms. The lowest-scoring images are smooth, closed pack lids, for which a Laplacian-variance metric reads flat metal as blurred, and the greyscale images originate entirely from two monochrome sources. Both categories add useful variation to the corpus, so deletion on the metric alone would degrade it.

Secondly, the harmful category is unlabelled images that nonetheless contain visible components. A proportion of the 23.5% is legitimate background imagery, which is beneficial in moderation because it suppresses false positives. However, inspection confirmed open packs with clearly visible busbar assemblies carrying no annotations at all. Such frames actively teach the detector that busbars are background, and they are a plausible contributor to the weak busbar accuracy reported in Section 5.5.

The audit also established that 16,945 of the 19,522 annotations are polygon masks, with between 5 and 151 vertices, rather than axis-aligned boxes. These are converted to enclosing boxes for detection training and evaluation. It should be noted that an early version of the evaluation pipeline silently discarded every polygon-format label, trained on 20% of the available annotations, and produced an inflated score that appeared plausible on inspection. The failure is easy to introduce and difficult to detect, and ground-truth instance counts should be reconciled across pipelines as a matter of routine.

3.4 The Cross-Facility Benchmark

A cross-facility benchmark of 225 images containing 780 instances was drawn from all sources and held out entirely from training. A convention- consistent subset of 177 images, obtained by applying the screening procedure of Section 4.3, is reported alongside the full benchmark throughout Section 5.

4 METHOD

4.1 Detectors and Training Configurations

Three configurations are compared.

The single-source specialist is a YOLOv8n detector trained only on the single-laboratory source at an input resolution of 768 px, following the conventional approach in this literature and reproducing the setting under which published accuracy figures for this task are typically obtained.

The multi-source YOLO11n configuration (Jocher and Qiu, 2024) comprises 2.6 M parameters and was trained from COCO initialisation (Lin et al., 2014) on all 13 sources for 150 epochs at 640 px with stochastic gradient descent.

The multi-source RF-DETR configuration (Roboflow, 2025) is a real-time detection transformer with a frozen DINOv2 backbone (Oquab et al., 2024), fine-tuned on the same data for 60 epochs at 384 px. The resolution and epoch budget are the architecture defaults published with the model; the consequence for comparability is stated in Section 7.

4.2 Harmonised Evaluation Protocol

Confidence scores are not comparable across architectures, and mixing metric implementations produced a spurious result in early experiments conducted for this study. Every model is therefore scored with a single metric implementation, at a common confidence threshold of 0.05, on identical images, against identical ground truth including polygon-derived boxes. The threshold of 0.05 was selected because it lies below the operating point of all three detectors and therefore does not truncate the precision-recall curve for any of them.

4.3 Convention-Consistency Screening

The proposed screening procedure identifies sources whose annotation convention diverges from the corpus consensus, using a single merged-corpus training run and no additional annotation.

Let $\mathcal{S} = \{S_1, \ldots, S_n\}$ denote the sources contributing to the benchmark, and let a_i denote the class-specific mAP@50 obtained by the merged-corpus detector on the held-out images of source S_i. A source is flagged as convention-divergent when its accuracy falls below a robust lower fence, which is defined as follows:

$$a_i < \mathrm{med}(\mathbf{a}) - k \cdot c \cdot \mathrm{mad}(\mathbf{a}), \tag{1}$$

where $\mathbf{a} = (a_1, \ldots, a_n)$ denotes the vector of per-source accuracies, med denotes the median, mad denotes the median absolute deviation, $c = 1.4826$ rescales the median absolute deviation to a consistent estimator of the standard deviation under normality (Rousseeuw and Croux, 1993), and k denotes a sensitivity parameter. A value of $k = 3$ is used throughout, which is the conventional choice for outlier rejection and which is shown in Section 5.2 to separate the divergent source from the consensus by a wide margin on this corpus.

The median and the median absolute deviation are used in place of the mean and the standard deviation because a divergent source inflates both of the latter and thereby partially masks itself. Consequently, the proposed procedure remains sensitive when the divergent source is large, which is the case on this corpus.

A source flagged by the proposed procedure is not discarded from the corpus. It is reported separately as an out-of-convention benchmark, on the grounds established in Section 5.2 that its labels encode a different task rather than a degraded version of the same one. The procedure requires no additional annotation, no additional training run beyond the merged-corpus model that is trained in any case, and no prior assumption about which source is divergent.

4.4 Training-Free Screening from Annotation Geometry

The procedure of Section 4.3 requires a merged-corpus training run. A cheaper variant operates on the annotation files alone, which permits a corpus to be screened before any detector is trained.

A source that annotates sub-components rather than pack-level modules produces many small boxes per image, whereas a source following the consensus convention produces few large ones. That signature is captured by a granularity index, which is defined as follows:

$$g_i = \frac{n_i}{s_i}, \tag{2}$$

Table 3: Cross-facility generalisation gap, mAP@50.

Model	In-domain	Cross-facility
Single-source specialist	0.818	0.277
Multi-source YOLO11n	0.195	0.410
Multi-source RF-DETR	0.541	0.502

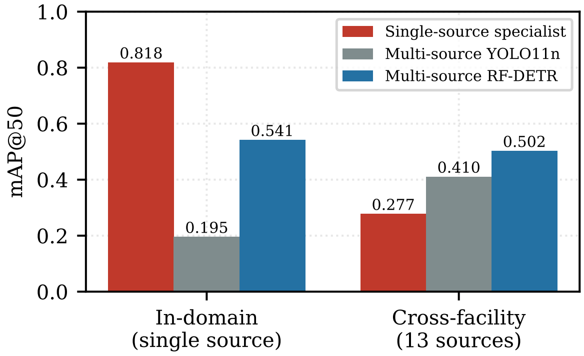

Figure 1: In-domain against cross-facility accuracy for the three configurations. The specialist attains the highest in-domain score and the lowest cross-facility score, and the ordering of the three models reverses between the two benchmarks.

where n_i denotes the mean number of module instances per image in source S_i and s_i denotes the median normalised module scale, taken as the square root of the enclosing-box area in normalised image coordinates. Scale is expressed as a square root so that the index is dimensionally a count per unit length and is therefore invariant to image resolution. The robust criterion of Equation 1 is then applied to $\mathbf{g}$ with the inequality reversed, since convention divergence raises the index rather than lowering it.

5 RESULTS

5.1 The Single-Source Evaluation Gap

Table 3 and Figure 1 quantify the central problem. The specialist attains 0.818 mAP@50 on the single-source benchmark on which it was trained and validated, and 0.277 on the cross-facility benchmark, a relative loss of 66%. Its in-domain accuracy therefore does not predict deployment behaviour.

The converse also holds and is worth stating: the multi-source models score lower in domain than the specialist. Training for breadth costs peak accuracy on any single pack type, and which trade is preferable depends on whether a line receives one pack type or many. However, the magnitude of the drop for YOLO11n, to 0.195, exceeds what breadth alone explains, and it is not a training failure. The in-domain benchmark is itself the source whose annotation convention diverges from the remainder of the corpus. That confound is examined in Section 5.2 and stated as a limitation in Section 7.

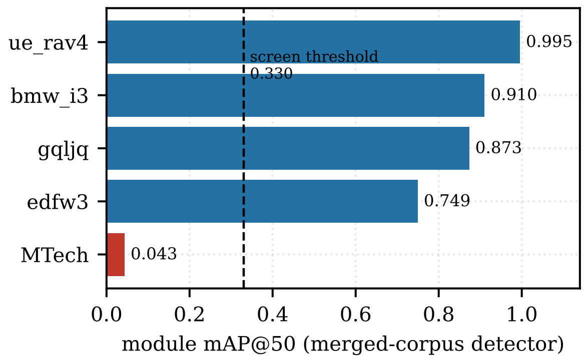

Figure 2: Per-source module accuracy for the merged-corpus detector, with the robust lower fence of Equation 1 evaluated at $k = 3$. One source falls below the fence by a wide margin.

Table 4: Sensitivity of the screening partition to the parameter k in Equation 1.

k	Lower fence	Sources flagged
1.0	0.786	2
1.5	0.710	1
3.0	0.330	1
4.5	−0.049	1
5.0	−0.176	0

5.2 Screening Outcome and Recovery

Per-source module accuracy for the merged-corpus detector is 0.995 on ue_rav4, 0.910 on bmw_i3, 0.873 on the largest source, 0.749 on ev-battery-components, and 0.043 on the single-laboratory source. Applying Equation 1 with $k = 3$ yields a lower fence of 0.330. Figure 2 illustrates that exactly one source falls below that fence, and that the margin is wide: the flagged source lies 0.287 below the fence, whereas the nearest consensus source lies 0.419 above it.

Table 4 presents the sensitivity of the partition to k. Any value between 1.5 and 4.5 produces the identical partition, so the outcome does not depend on a tuned threshold. Below $k = 1.4$ the procedure additionally flags a consensus source, and above $k = 4.6$ it flags nothing.

Visual inspection of the flagged source confirms the diagnosis. It annotates small internal sub-components in extreme macro views as modules, whereas every other source annotates pack-level modules. The detector learned the majority convention and applied it consistently, so the collapse constitutes a disagreement about the target rather than a failure of perception.

The consequence is quantified in Figure 3 and Table 7. Excluding the flagged source from the benchmark raises YOLO11n module mAP@50 from 0.547 to 0.774, a recovery of 0.227, with no additional data

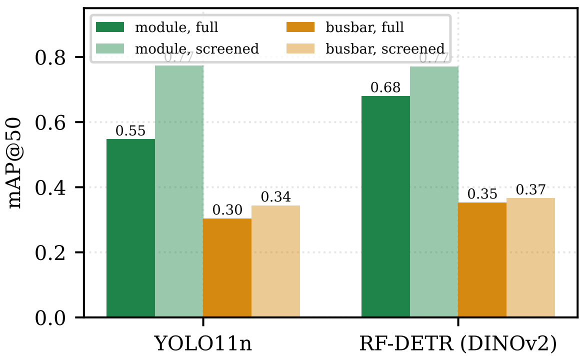

Figure 3: Per-class accuracy on the full benchmark and on the screened subset. Screening recovers 0.227 module mAP@50 for YOLO11n and 0.091 for RF-DETR, and leaves busbar accuracy substantially unchanged.

and no additional training budget. Aggregate accuracy computed over sources with inconsistent conventions is therefore not a meaningful quantity, and per-source reporting is necessary rather than merely informative.

5.3 Agreement of the Training-Free Screen

Equation 2 was evaluated over all 4,760 label files in the corpus, using no images and no trained model. Figure 4 presents the outcome. The divergent source attains a granularity index of 40.7, against 15.0 for the next highest source and a median of 9.00 across the corpus. Applying the robust criterion with $k = 3$ yields an upper fence of 29.02, and exactly one source exceeds it.

The source identified is the same one flagged by the model-based procedure of Section 5.2. The two screens are methodologically independent: one measures how a trained detector behaves on held-out images of each source, and the other measures only the geometry of the annotations themselves, with no model involved. Their agreement provides a second and independent line of evidence that the source encodes a different annotation convention rather than presenting harder imagery.

The practical implication is that a corpus may be screened before a detector is trained. It should be noted that the training-free index detects divergence in annotation granularity specifically, which is the form of divergence present in this corpus. Conventions that disagree about class boundaries while preserving object scale, for example a source that labels a cooling plate as a module, would not be separated by Equation 2 and would still require the model-based procedure.

5.4 Positive Control

Both screens were applied to a corpus in which the divergent source was identified after the fact, so neither has been tested against a case whose answer is known in advance. To supply one, a controlled positive was constructed. A consensus source was relabelled to a deliberately different convention, in which every pack-level module box is replaced by a 3×2 grid of inset sub-component boxes covering the same region, imitating the granularity of the real divergent source. No image is altered and no busbar annotation is touched, so any screen responding to the manipulation responds to annotation convention and to nothing else.

Table 5: Positive control. A consensus source is relabelled to a sub-component convention; granularity index before and after, with the $k = 3$ fence.

Source	Before	After	Flagged
manipulated (consensus)	11.1	185.8	after only
real divergent source	40.7	40.7	both
next consensus source	15.0	15.0	neither
fence ($k = 3$)	29.02	35.24	

Table 5 presents the outcome. The manipulated source carries a granularity index of 11.1 before manipulation, which places it inside the consensus band, and 185.8 after. The training-free screen flags it, continues to flag the real divergent source, and leaves every genuine consensus source below the fence. The procedure therefore recovers a divergence whose ground truth is known by construction, without producing false positives on the remaining sources.

It should be noted that this control validates the training-free screen of Section 4.4. An equivalent control for the model-based procedure of Section 4.3 requires a detector to be retrained on the manipulated corpus, which is reported as future work. The manipulation script is released with the benchmark so that the control can be reproduced and extended.

5.5 Architecture Comparison under Annotation Shift

Table 6: Detector comparison on the cross-facility benchmark of 225 images.

Model	mAP@50	mAP@50–95	module	busbar
Specialist	0.277	—	0.231	—
YOLO11n	0.410	0.256	0.547	0.304
RF-DETR	0.502	0.312	0.680	0.353

Under the harmonised protocol, RF-DETR attains 0.502 mAP@50 against 0.410 for YOLO11n on the full benchmark, and leads on the stricter mAP@50–95, at 0.312 against 0.256. Table 6 presents the full per-class breakdown.

Taken alone that comparison is misleading. Table 7 reports the same two models on the screened subset of 177 images, on which they are effectively tied: 0.558 against 0.544 overall, 0.771 against 0.774 on the module class, and 0.366 against 0.344 on the busbar class. The frozen self-supervised backbone therefore confers

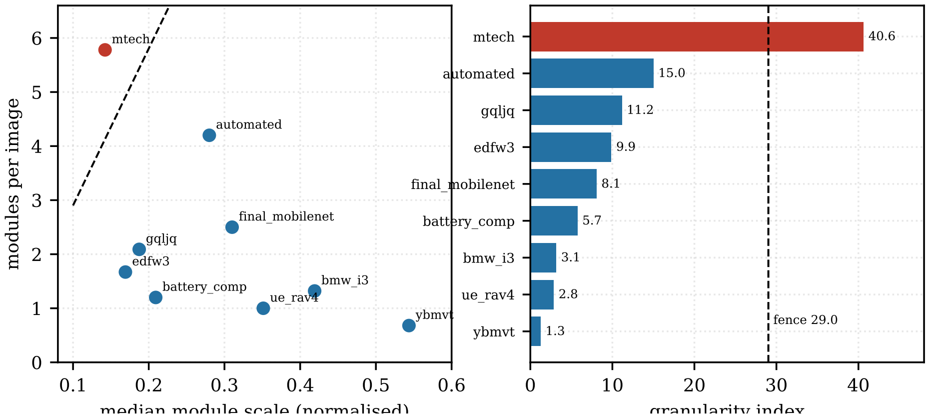

Figure 4: Training-free convention screening from annotation geometry. Left: median module scale against modules per image for every source contributing at least 40 module instances; the dashed line is the screen fence. Right: the granularity index of Equation 2 with the robust fence at $k = 3$. The source flagged here is the same source flagged by the model-based procedure of Section 4.3, and it is identified without any training run.

Table 7: Effect of screening on the architecture comparison, mAP@50. The final row reports the module accuracy lost when the flagged source is reintroduced.

	YOLO11n		RF-DETR	
Benchmark	overall	module	overall	module
Screened (177)	0.544	0.774	0.558	0.771
Full (225)	0.410	0.547	0.502	0.680
Module loss	−0.227		−0.091	

no advantage on conventionally annotated packs.

The aggregate advantage therefore arises entirely from behaviour under annotation shift. Reintroducing the flagged source reduces YOLO11n module accuracy from 0.774 to 0.547, a loss of 0.227, but reduces RF-DETR module accuracy only from 0.771 to 0.680, a loss of 0.091. The defensible claim is correspondingly narrow: a frozen self-supervised backbone does not detect ordinary modules more accurately, but it degrades considerably more gracefully when the definition of the target moves. On a line receiving heterogeneous packs, graceful degradation is the property of practical value, and the proposed screening procedure and the choice of backbone are therefore complementary rather than alternative responses to the same problem.

Busbar is the weaker class for every configuration, at 0.30 to 0.37. The cause is annotation coverage rather than model capability: only four of eight benchmark sources annotate busbars at all, and 61% of busbar instances originate from a single source. This is the same coverage problem identified for the module class, expressed in a different form, and it indicates that broadening busbar annotation across sources is a more productive intervention than architectural change.

5.6 Zero-Shot Transfer to Unseen Pack Families

To characterise transfer beyond the corpus, the merged-corpus detector was additionally evaluated on 17 pack families absent from it, drawn from an unlabelled public collection. Because no annotations are available for that collection, a detection-rate proxy is reported rather than mAP.

The mean detection rate is 0.76 and busbars are found in 16 of the 17 families. Figure 5 illustrates the spread. Accuracy is high on saloon and hatchback packs, at 1.00 on BMW i4 and Hyundai Ioniq and 0.95 on Ford Mondeo, and low on geometries unlike anything present in the corpus, at 0.40 on Mercedes GLE and 0.27 on a Volvo truck pack. Module geometry therefore transfers to unseen passenger-vehicle packs, whereas unusual form factors remain the weak axis. It should be noted that a detection-rate proxy measures recall only and cannot detect false positives, so this result is corroborating evidence rather than a substitute for the labelled benchmark.

5.7 Deployment Cost

Dynamic INT8 quantisation of the detector to the ONNX format reduced median CPU latency from 56.3 ms to 44.8 ms and file size from 6.3 MB to 3.4 MB, for a loss of 0.007 mAP@50. Latency was measured as the median over 20 benchmark images on an Apple M1 processor with no graphics acceleration. The detector is therefore deployable without a graphics processor at approximately 22 frames per second, which is sufficient for the inspection rates typical of a manual disassembly station.

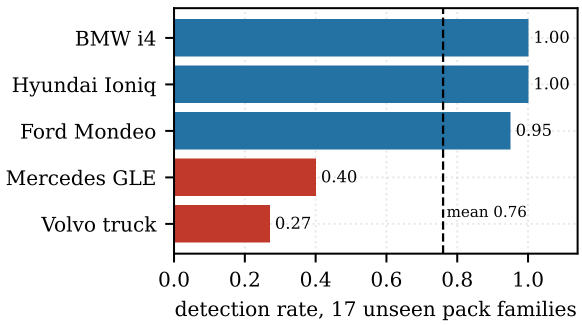

Figure 5: Zero-shot detection rate on pack families absent from the corpus. Accuracy is high on passenger-vehicle packs and degrades on form factors unlike anything present in training.

5.8 Negative Results

Four failures constrain the design space for this task. Each is reported because it was expected to succeed and because the expectation is common in the surrounding literature.

Open-vocabulary detection is unusable for this task. YOLO-World (Cheng et al., 2024), prompted with ev battery module and busbar, attained 0.004 mAP@50 on the benchmark. Battery internals lie too far outside the vision-language pre-training distribution for the prompts to ground, so the annotation-free route is not currently available for this application.

Naive multi-source merging degrades accuracy. Training on the merged corpus without screening reduced module mAP@50 to 0.094 and aggregate accuracy to 0.331 on the single-source test split, against a single-source baseline of 0.818. More data made the model worse. This result motivated the screening procedure of Section 4.3 and illustrates that curation dominates volume when conventions conflict.

Automatic relabelling cannot reconcile a divergent convention. Reconciliation of the flagged source with the consensus convention was attempted using Grounding DINO (Liu et al., 2024). On clean frames the model recovered two of approximately five modules, and on the greyscale macro imagery that dominates the source it produced boxes covering half the frame. Relabelling 1,756 images on that basis would have injected substantial noise. Convention conflicts are therefore not cheaply repairable after collection, which argues for agreeing annotation guidelines before collection rather than after.

Copy-paste and glare augmentation degrade the detector. Copy-paste compositing (Ghiasi et al., 2021) combined with a synthetic glare pass was applied in order to address the specular highlights characteristic of battery internals. Validation mAP@50 declined monotonically over the first six epochs, from 0.503 to 0.270 against a baseline of 0.818, and training was terminated. The likely cause is label noise introduced at paste boundaries together with glare intensity that does not match the statistics of the real bright-condition imagery. Both techniques are widely reported as beneficial elsewhere, which indicates that their transfer to this domain should not be assumed.

6 DISCUSSION

Three practical implications follow from the results presented above.

Firstly, accuracy reported on a single-source benchmark should not be treated as an estimate of deployment accuracy for this task, and the gap is large enough to change engineering decisions rather than merely to qualify them. A detector selected on in-domain accuracy alone would have selected the specialist, which is the worst of the three configurations on the cross-facility benchmark.

Secondly, when a corpus is assembled from heterogeneous sources, per-source disaggregated reporting is a requirement rather than an option. An aggregate figure computed across conventions that disagree describes no realisable deployment condition. The proposed screening procedure makes the necessary partition explicit and reproducible, at the cost of a single evaluation pass over a model that is trained in any case.

Thirdly, the choice of backbone and the curation of the corpus address the same failure mode by different means, and the results indicate that curation is the more effective of the two. Screening recovers 0.227 module mAP@50 for YOLO11n, whereas substituting a frozen self-supervised backbone recovers 0.133 of that same loss at substantially greater inference cost. The two are nevertheless complementary, since a divergent source that has not been identified is still handled more gracefully by the self-supervised backbone.

7 LIMITATIONS

The 66% relative loss reported in Section 5.1 is measured against an in-domain benchmark that is itself the convention-divergent source. The figure therefore conflates generalisation loss with the specialist having been trained on an idiosyncratic convention, and it should be read as an upper bound on the generalisation component rather than as a pure measurement of it. A second in-domain reference source would separate the two contributions.

Each configuration was trained once, so no seed variance is available and the 0.092 mAP@50 margin between RF-DETR and YOLO11n is not accompanied by a confidence interval. The two configurations also differ in epoch budget, input resolution and optimiser, because the architecture defaults were used in each case. The comparison should accordingly be read as a comparison of two configurations as they are commonly deployed, and not as a controlled comparison of two architectures.

The benchmark inherits the annotation conventions of its sources and no independently relabelled gold subset was constructed, so model error and label error cannot be separated on it. This is a material limitation for a study whose central claim concerns annotation quality, and its removal is the first item of future work.

The proposed screening procedure requires that the divergent sources are a minority of the corpus, since the median defines the consensus. It has been validated on a corpus containing one divergent source and would require modification for a corpus in which two or more mutually incompatible conventions are each well represented.

The corpus is assembled from public sources rather than collected under controlled conditions, so pack-type coverage is uneven and licences vary. One source representing 24% of the training images is licensed CC BY-NC-SA 4.0, whose ShareAlike condition propagates to any derivative corpus.

Finally, absolute accuracy remains modest. Busbar accuracy of 0.35 is not sufficient for autonomous operation on that class, and the results presented here characterise the failure modes of the task rather than deliver a deployable detector.

8 CONCLUSIONS

This paper presented a cross-facility benchmark for electric vehicle battery component detection and used it to establish three results. Single-source evaluation overstates deployment accuracy sufficiently to mislead: the same detector lost two thirds of its accuracy when moved across pack sources. Annotation convention, rather than architecture, proved the dominant failure mode, and the proposed convention-consistency screening procedure identified the divergent source from a single training run and recovered 0.227 module mAP@50 by excluding it. A frozen self-supervised backbone was shown to supply robustness to annotation shift rather than improved detection of ordinary modules, which narrows the circumstances under which it should be preferred.

Taken together, these results indicate that dataset design, and specifically agreement on annotation convention before collection, is the first thing to get right in this application. A training-free variant operating on annotation geometry alone was shown to flag the same source, which permits a corpus to be screened before any detector is trained.

Future work will construct an independently relabelled gold subset in order to separate model error from label error, will extend the positive control to the model-based screen by retraining on the manipulated corpus, and will extend the training-free index to forms of convention divergence that preserve object scale.

DATA AVAILABILITY

The benchmark definition, the per-source provenance and licence record, the dataset audit tooling, the evaluation code and the trained detector weights are available at https://github.com/Komil-jon/ev-battery-vision-pipeline.

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